

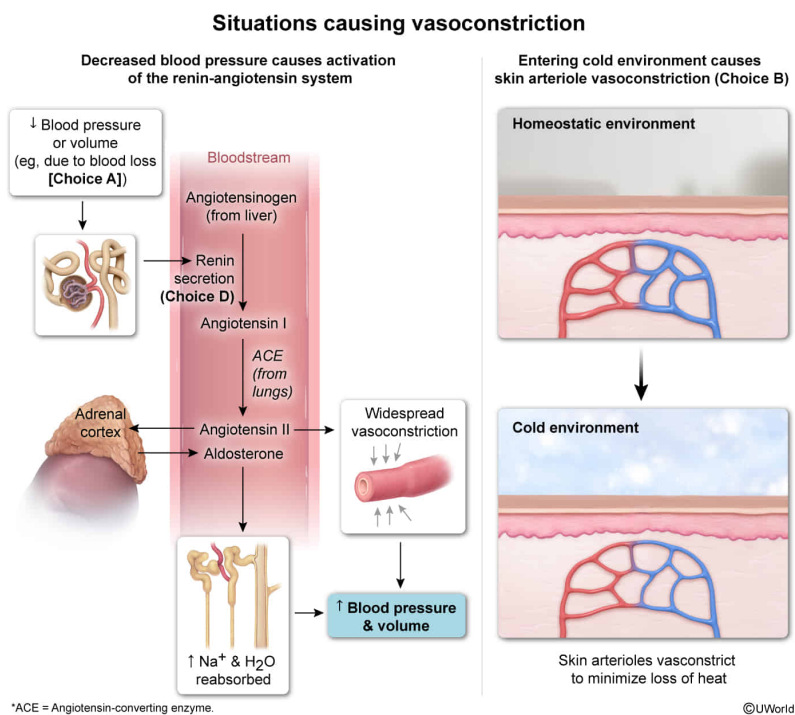
Which of the following situations would NOT cause vasoconstriction to occur?

- ☐ A. Loss of blood [15%]
- ☐ B. Exposure to cold temperatures [5%]
- ✓ ☒ C. Increased blood pressure [61%]
- ☐ D. Secretion of renin [17%]

Correct

60% answered correctly

Explanation:



Vasoconstriction and vasodilation (ie, manipulation of **blood vessel diameter**) are mechanisms that maintain blood pressure and blood volume within normal ranges, preserving **homeostasis**. **Blood vessels** are surrounded by **smooth muscle**, which can contract or relax, changing their diameter. This directly affects the pressure of the blood flowing through the vessels.

During **vasoconstriction**, the smooth muscle surrounding the vessel contracts. This results in a decrease in vessel diameter, which consequently increases the resistance on blood flowing through the vessel. This increased resistance leads to an increase in blood pressure. For example, if blood pressure were to decrease, homeostatic mechanisms would cause vasoconstriction to occur, reversing this decrease and leading to an increase in blood pressure, restoring homeostasis.

Conversely, during vasodilation, the smooth muscle surrounding the vessel relaxes. This results in an increase in vessel diameter, leading to a decrease in blood pressure due to decreased resistance to blood flow. For example, if blood pressure were to increase, homeostatic mechanisms would cause vasodilation to occur, reversing the increase and leading to a decrease in blood pressure, restoring homeostasis.

Examples of situations in which vasoconstriction *does* occur include:

- **(Choice A)** Blood loss (eg, due to injury) can severely decrease blood volume and thus blood pressure. Vasoconstriction reverses this by narrowing vessel diameter and increasing the pressure of remaining blood.
- **(Choice B)** The skin is involved in the maintenance of **body temperature homeostasis**. When exposed to cold temperatures, skin arterioles *constrict* to minimize loss of body heat by diverting warm blood from skin capillaries to the body's core.
- **(Choice D)** In response to a decrease in blood pressure or blood volume, **renin** is secreted by the kidney to

promote vasoconstriction and increase blood pressure.

(Choice C) The only scenario in which vasoconstriction would *not* occur would be an increase in blood pressure. In this case, vasoconstriction would cause the blood pressure to increase to an even greater extent, taking the individual *further* from homeostasis. Therefore, an increase in blood pressure would lead to vasodilation, which decreases blood pressure to homeostasis levels.

Educational objective:

When blood pressure is low, vasoconstriction reduces blood vessel diameter, causing an increase in blood pressure to restore homeostasis.

Time spent:0 sec

Subject:

Biology

QID:403142

System:

Circulation and Respiration